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Roll No

MI-3002-CBGS

B.E., III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mining Environment - I

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is relative humidity?
b) Write short notes on:
 - i) Hygrometer
 - ii) Kata Thermometer
 - iii) Aneroid Barometer
2. a) Explain Coward's Diagram.
b) Define:
 - i) Laminar flow and Turbulent flow
 - ii) Losses in mine (ventilation)
3. a) Discuss the effects of heat and humidity in mines.
b) Briefly explain the psychometric properties of mine air.

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4.
 - a) Describe the mine illumination systems.
 - b) Write the safety features of mine lamp also discuss its maintenance.
5.
 - a) Write a detailed note on control of heat and humidity in mines.
 - b) Explain Square law of mine ventilation.
6. Explain:
 - a) Detection of carbon monoxide (CO) by P.S. detector.
 - b) Anemometer
7.
 - a) Explain (Any one)
 - i) MSA D6 Methanometer
 - ii) Infra-red gas analysers
 - b) What do you understand by Methane drainage? Explain all the methods of methane drainage.
8. Explain the standards of illumination for different places in underground and open cast mines.

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